Laboratory Activity: Queueing Models and Applications

Course: Modeling and Simulation / Operations Research
Week 11 – BSCS Laboratory
Duration: 2–3 hours
Prepared by: **Daniel G. De Guzman**

I. Objectives

After completing this lab, BSCS students will be able to:

1. Model and simulate single-server and multi-server queues using real-life problems.
2. Apply Markov Chain concepts to simulate state-based transitions.
3. Develop simulation code using Python for queue analysis.
4. Evaluate system performance (waiting time, queue length, utilization).
5. Interpret how queueing theory helps in designing efficient real-world systems.

II. Tools and Environment

- Python (Jupyter Notebook or PyCharm)
- Libraries: `random`, `numpy`, `matplotlib`, `simpy` (optional)
- Real-life scenarios provided as case prompts

III. Lab Activities

◆ Task 1: Conceptual Review (15 minutes)

Answer the following:

1. What is the primary difference between M/M/1 and M/M/c queueing models?
2. Why do businesses prefer multi-server models in high-traffic scenarios?
3. What is a Markov Chain, and how does it apply to a queueing system?

◆ Task 2: Scenario 1 – Bank Queue Simulation (M/M/1) (45 minutes)

Scenario:
You are designing a simulation for a **local bank** that has one teller during lunch hours. On average, **10 customers arrive per hour**, and the teller serves customers at an average rate of **12 per hour**.

Tasks:

- Simulate 50 customer arrivals.
- For each customer:
  - Generate interarrival time and service time (exponentially distributed).
  - Calculate arrival, service start/end, and waiting time.
- Output:
  - Average wait time
  - Average queue length
  - Server (teller) utilization

📈: Plot a Gantt chart showing service usage over time.

◆ Task 3: Scenario 2 – Fast-Food Restaurant with Multiple Counters (M/M/3) (45 minutes)

Scenario:
A popular **fast-food chain** opens during peak hours with **3 counters** (servers). On average, **30 customers arrive per hour**, and each server can serve at a rate of **12 customers/hour**.

Tasks:

- Modify your code to simulate a **multi-server** setup.
- Assign each arriving customer to the next available counter.
- Output:
  - Average waiting time
  - Queue length over time
  - Server utilizations (each counter)

□ **Reflection**: How did adding more servers affect system performance compared to the bank?

◆ Task 4: Scenario 3 – Traffic Light Control with Markov Chains (30 minutes)

Scenario:
You are developing a **traffic simulation system** for an intersection with a single light that cycles through **Green → Yellow → Red**. Each state change follows a probability model.

Markov Chain Setup:

- Green (G) → Yellow (Y) with P=1.0
- Yellow (Y) → Red (R) with P=1.0
- Red (R) → Green (G) with P=1.0

Extended Behavior:

- Cars arrive every 5–10 seconds.
- Red light causes cars to queue.
- Green light releases 1 car every 2 seconds.

Tasks:

- Simulate 60 light cycles.
- Track:
  - Car queue size per cycle
  - Wait time for cars during red lights
- Plot average car wait time and queue size over simulation

IV. Submission Requirements

Submit a **.zip** file containing:

1. Python source code (`.py` or `.ipynb`)
2. Output screenshots or graphs (PNG, JPG)
3. Written report:
   - Answers to Task 1
   - Analysis and comparison of Task 2 and Task 3
   - Summary insights from Task 4 simulation

V. Grading Rubric

Criteria	Points
Task 1 Conceptual Answers	10
Task 2: M/M/1 Bank Simulation	20
Task 3: M/M/3 Restaurant Simulation	20
Task 4: Traffic Light with Markov	20
Code Quality & Documentation	10
Graphs and Visuals	10
Report Reflection and Insights	10
Total	**100**